Tutorial 3 - SPM (wk 8 - done)

Question 1 (Team 1)

Refer to the case study in Tutorial 2 Question 1 (Beamscope company),
Suggest **ONE (1)** *software quality attribute* for Beamscope's project implementation. Explain your answer.
(Refer to Appendix 3.1)

Suggested Software Quality Attribute: Performance (Efficiency)

Efficiency refers to the ability of the system to perform its functions quickly and with minimal use of system resources such as time, memory, or processing power. A highly efficient system responds promptly to user requests and processes large volumes of data or tasks without unnecessary delays.

Reason:

1. **To Reduce Order Processing Time**

In the old system, it took 5 to 7 days to ship an order due to manual steps like handwriting orders, checking credit manually, and confirming stock by phone. This caused serious delays and customer dissatisfaction. So that, The new system should allow real-time order placement, instant inventory checking and automated processing. This dramatically improves speed, ensuring faster order fulfillment and a better customer experience.

2. **To Maximize Operational Productivity**

Under the old manual system, staff had to spend a significant amount of time on repetitive and low-value tasks such as searching for inventory, confirming orders by phone, and compiling paper-based reports. These processes not only slowed down operations but also increased the risk of human error. With the new system, many of these tasks are automated—inventory is tracked using bar-code and radio frequency technology, order data is centralized, and reports are generated in real time. This reduces manual workload, minimizes errors, and allows employees to focus on more strategic tasks, ultimately increasing the company's overall efficiency and capacity to handle larger volumes of business with fewer resources.

Question 2

Team 2

Question 2

Quality Assurance (QA) and Quality Control (QC) activities are important in software quality management.

Identify and explain **THREE (3)** differences between Quality Assurance (QA) and Quality Control (QC) in software development.

- Software Quality Management (SQM)'s 3 main activities:
  1. Quality Assurance (QA): to develop an organizational framework (procedures & standards) that will lead to high quality of software.
  2. Quality Planning (QP): to select appropriate procedures and standards from the framework and adapt to a specific software project.
  3. Quality Control (QC): to execute processes to ensure that software development follows the quality procedures and standards.

Aspect	Quality Assurance (QA)	Quality Control (QC)	Example
Focus	QA focuses on preventing defects by improving development processes and standards.	QC focuses on detecting defects in the actual software product through testing.	QA : Implementing coding standards to avoid bugs. QC : Running unit tests to find bugs.
Approach and Timing	QA is **proactive** and done throughout the software development lifecycle	QC is **reactive** and done after the product is developed (during or after testing)	QA : Conducting process audits during development. QC : Performing system testing before release.
Responsibility	QA is the responsibility of all team members, especially those involved in the process.	QC is mainly the responsibility of the testing team or quality control engineers.	QA : A developer follows peer review practices QC : Testers execute test cases to find bugs.

Question 3
Team 3
Question 3
Errors and defects found in a software will reduce the software quality. However, there are many techniques to avoid errors and defects in software projects. Below are the two of them:
- Inspections - Lessons learnt report

Describe each of the two above techniques.

1. Inspections

- A **formal review process** where project artifacts (code, design documents, etc.) are systematically examined by a team.
- Aim: Detect and fix defects early in development (before testing phase).
- Example: Code inspections before integration to catch logic errors, style violations, o security flaws.

- **refer slide pg30**

2. Lessons Learnt Report

- A document created after the completion of a project or project phase.
- It records **successes, mistakes, and recommendations** for future projects.
- Helps teams **avoid repeating past errors** and improve quality continuously.

- **records What when well**
- **Created by project manager**

****Q3 add in gantt chart**

Question 4
The president of ABC College has recently assigned you to manage a new project that is to develop an Access Card Control System for enhancing the security in the campus. For implementing this system, the quality of the software is as important as the quality of hardware such as RFID devices. In order to produce good quality software, it is necessary to

perform quality management activities. However, performing quality management activities will incur costs.

Describe **3 categories** of quality costs as given below:

- Prevention costs **- Appraisal costs** **- Failure costs**

Explain which category is likely to be most costly to the ABC College. Justify your answer.

Answer:

Prevention costs	**Appraisal costs (Checking)**	**Failure Costs**
1The cost of management activities - to plan and coordinate all Quality assurance(QA) and Quality control(QC)	The cost of conducting technical reviews.	**Internal failure costs** ● Costs to correct design error ● Cost to correct program error Internal failure cost is a cost incurred before product delivery
The cost of technical activities - to develop complete requirements and design models Example: Training the employees to use Access Card Control System developed, to prevent errors of using in the future.	**Cost of testing products.** - Designing new test cases. - Setting up a testing environment. - Carrying out new test executions	**External failure costs** ● Resolving customer complaints ● Handling product return and replacement ● Providing help line support ● Managing a poor reputation This is the cost incurred after the product has been delivered to customers
-	**Cost of collecting data and reporting inspection.** Collecting reports as a result of data. Reporting the inspection for the closing criteria.	-
The goal is to ensure a project is error-free or within an acceptable range.	To ensure that a project is error-free or within an acceptable range.	To ensure the project is error-free, and the customer is satisfied with the solution provided

It is **failure costs,** if the Access Card Control System has defects after deployment such as allowing unauthorized access or failing to read RFID cards properly it could compromise campus security, damage the college's reputation, and require urgent and costly repairs. Moreover, handling complaints and possibly replacing hardware or software components adds substantial cost.

Internal Failure cost could be unauthorized access or failing to read RFID cards properly. This could cause damage to the college's reputation due to low quality security measures. In terms of External Failure Costs, it could be that clients requiring urgent and costly repairs to be done. This could add substantial cost by handling complaints, and possibly replacing hardware or software components.